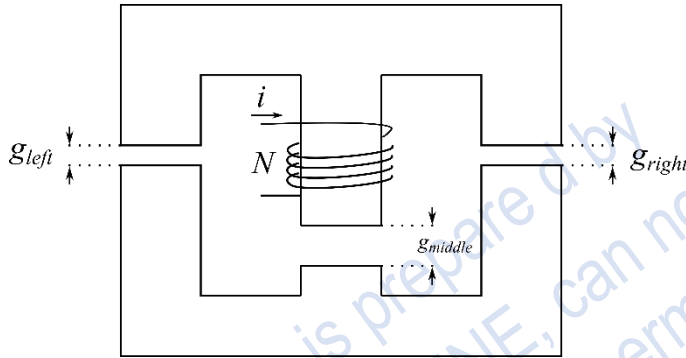


HOMEWORK 1



An E-core with three air-gaps is given in the figure with dimensions of left mean core length 10 cm, right mean core length 10 cm, middle leg mean core length of 4 cm and core cross section of 1 cm² and uniform through the geometry except 2 cm² at middle leg.

- i) Core is made out of a low-quality material "A" with relative permeability of 1000. The coil with $N = 300$ turns is carrying a current of 1 Amps.

a) Calculate the coil inductance when there are no air-gaps.

Left air gap is equal to the right air gap and **when $g_{\text{left}} = g_{\text{right}} = x$ cm, $g_{\text{middle}} = x + 0.1$ cm.** While left and right air gaps are simultaneously changed between $0 \text{ cm} \leq x \leq 0.1 \text{ cm}$ (use at least 100 data points):

- b) Calculate and plot inductance (L) of the coil with respect to middle gap length. What is the reason behind this change? Give your comments.
 c) Calculate and plot the middle and left air gap flux density with respect to respective air gap lengths. Comment on your graphs.

- ii) Now assume the core is made out of a high-quality material "B" with a relative permeability of 15000. Repeat the parts above. Comment on your findings. What is the reason of change, or lack thereof? How does adding an air-gap to a ferrous core affect the inductance?

Neglect leakage fluxes and fringing. Assume a uniform distribution of magnetic field inside the core.
 ($\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$)